

Bioprinting of Multiscaled Hepatic Lobules within a Highly Vascularized Construct

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Highly vascularized complex liver tissue is generally divided into lobes, lobules, hepatocytes, and sinusoids, which can be viewed under different types of lens from the micro- to macro-scale. To engineer multiscaled heterogeneous tissues, a sophisticated and rapid tissue engineering approach is required, such as advanced 3D bioprinting. In this study, a preset extrusion bioprinting technique, which can create heterogeneous, multicellular, and multimaterial structures simultaneously, is utilized for creating a hepatic lobule (≈ 1 mm) array. The fabricated hepatic lobules include hepatic cells, endothelial cells, and a lumen. The endothelial cells surround the hepatic cells, the exterior of the lobules, the lumen, and finally, become interconnected with each other. Compared to hepatic cell/endothelial cell mixtures, the fabricated hepatic lobule shows higher albumin secretion, urea production, and albumin, MRP2, and CD31 protein levels, as well as, cytochrome P450 enzyme activity. It is found that each cell type with spatial cell patterning in bioink accelerates cellular organization, which could preserve structural integrity and improve cellular functions. In conclusion, preset extruded hepatic lobules within a highly vascularized construct are successfully constructed, enabling both micro- and macro-scale tissue fabrication, which can support the creation of large 3D tissue constructs for multiscale tissue engineering.

1. Introduction

The human body is composed of various cell types that together create heterogeneous tissues and subsequently organ systems.^[1] This complex system can be considered as multiscaled tissues and organs. To engineer multiscaled heterogeneous tissues and organs, various tissue engineering approaches have been introduced,^[2] specifically 3D bioprinting.^[3–12] Extrusion-based

bioprinting is a promising branch of 3D bioprinting in terms of ease of use, low price, and extensive material selectivity compared to other bioprinting branches (i.e., inkjet bioprinting, laser-assist bioprinting).^[13] However, the limited printing resolution ($100\ \mu\text{m}$)^[14] is still a bottleneck in the engineering of microscale tissue constructs. Our previous study reported that a preset extrusion bioprinting technique could simultaneously improve printing resolution and cell viability.^[15] Likewise, recent advanced bioprinting technologies have enabled both multimaterial and multilayer printing under physiological conditions, enabling fabrication of useful medical devices easily.^[16,10,17] Although these functional and microscale tissue constructs can be achieved by advanced bioprinting technology, in vitro fabrication of these tissues is an intermediate step toward achieving organ-level complexity.^[18]

Because tissue engineering aims to restore tissue function by repair or regeneration of the damaged tissue,^[19] functional and microscale scale tissue

constructs should be fused together or directly printed to create large 3D tissue constructs. However, the size of an engineered tissue construct is limited to the cellular diffusional limit of oxygen and nutrients ($100\text{--}200\ \mu\text{m}$).^[20] In the tissue engineering field, it has been demonstrated that creation of cell-laden constructs without microchannels decreased cell viability and cellular differentiation compared to pre-existing microchannels.^[21] Thus, to avoid ischemic conditions, many researchers have utilized sacrificial materials to create microchannels^[21,22,7,23,11] or coaxial nozzles for in situ creation of blood vessels.^[24–26] Further, since capillaries are too small to be printed, a predefined experimental design is essential for interconnection of the microchannels during angiogenesis.^[27] As a result, to create large tissue constructs, both microchannels and a predefined design for angiogenesis are needed.

The human liver is a highly vascularized organ containing multiple vascular systems. A hepatic lobule (≈ 1 mm)^[28] is a small functional unit of the liver, which is composed of the sinusoidal lumen, Disse space, biliary canaliculi, hepatocytes, nonhepatocytes, sinusoidal endothelial cells, Kupffer cells, and stellate cells within the vascular system.^[29] This highly

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vascularized liver can be regarded as one of the representative multiscale tissues. Therefore, advanced bioprinting technology could be a key tool to fabricate thicker liver models for multiscale tissue fabrication. However, despite hepatocytes comprising 80% of the total liver volume, primary hepatocyte cultivation is still difficult under 2D conditions.^[30,31] To overcome this, 3D hepatocyte spheroid^[32,33] or 2D sandwich culture^[34,35] methods have been utilized for hepatotoxicity testing. A recent study proved that 3D hepatocyte spheroids were more functionally stable and exhibited increased sensitivity after long-term repeated exposures.^[36] Although the 3D hepatocyte spheroid is an ideal model of hepatotoxicity, the limited size of spheroids due to hypoxic conditions at the spheroidal core is an impediment for the creation of thicker liver tissues.

Further, there are several commercial suppliers producing liver tissue models and conducting research in the bioprinting area or 3D culture system, such as Organovo, InSphero, and Cyfuse.^[37] Organovo can print cell spheroids, which were prepared from cell pellet within capillary micropipettes, and fused together to create either multicellular spheroids or cylinders.^[38] From this, Organovo has created bioprinted 3D liver tissues composed of liver parenchymal/nonparenchymal cells, which enabled a vascular network of endothelial cells (ECs) and presence of quiescent hepatic stellate cells.^[39] Likewise, Cyfuse can print spheroids onto needle arrays for spheroid fusion, which is called the Kenzan method.^[40] A recent study from Cyfuse showed that bioprinted liver tissue composed of primary human hepatocytes and mouse fibroblasts could maintain functional metabolism of drugs, glucose, lipids, and bile acid for long periods.^[41] These approaches, while promising, are based on spheroid culture and post-bioprinting. Therefore, simplification of the printing procedure is needed for high-throughput analysis or batch-to-batch variation. Instead, InSphero showed a platform that extends the hanging-drop technology, used for multicellular spheroid formation, to multifunctional complex microfluidic networks.^[42] One of these liver platforms, including a diet-induced steatosis model and nonalcoholic steatohepatitis models,^[43] can be used for hepatotoxicity studies and predictive screens of new compounds.^[44] Although multiorgan-on-a-chip techniques provided great benefits in the simulation of human bodily responses, bottom-up tissue engineering with a combination of cell-biomaterials constructs for scaling up microtissues or organoids remains challenging.^[45]

Taken together, development of a bioprinting technique has the capability to fabricate vascularized multiscale tissue constructs. These bioprinting techniques can be used in conjunction with other emerging biotechnologies to aid development of alternative animal testing and contribute to the success of clinical trials. In this study, we successfully printed a hepatic lobule structure that embeds a high cell density of HepG2/C3A and ECs with a central vein using a preset cartridge. Specifically, ECs were well organized by covering the exterior of the construct, forming an endothelium-lined lumen and interconnecting between the exterior and the lumen, which enabled maintenance of the structure and cell survival. The engineered hepatic lobule constructs showed higher albumin and urea secretion compared to a nonengineered group, including a monocultured model in 2D conditions. Further, protein expression levels of albumin, MRP2, and CD31 as well as enzyme

activity of CYP3A4 and CYP1A1 were higher. Finally, we achieved the multiscale hepatic lobe construct (e.g., hepatic lobule arrays) using layer-by-layer bioprinting.

2. Results and Discussion

2.1. Principle of Hepatic Lobule Printing

We previously reported that the preset extrusion bioprinting technique could create heterogeneous, multicellular, and multimaterial structures simultaneously, depending on the design of the preset cartridge.^[15] In this study, we designed and fabricated a new preset cartridge that resembles the structure of a human hepatic lobule (**Figure 1**). A hepatic lobule is a building block of the liver parenchyma, which is composed of hepatocytes arranged in the liver cords within a sinusoid network and a central vein.^[29] Thus, to engineer this complex hepatic lobular structure, the preset cartridge was fabricated with three individual segments to preserve hepatocyte-laden bioinks, ECs-laden bioinks, and cell-free bioinks as sacrificial materials. To visualize this, we embedded fluorescent particles containing gelatin solution (blue, sacrificial component) and alginate solutions (green, red) into the preset cartridge and printed as shown in the schematic illustration (**Figure 1A,C**). Subsequently, the alginate bioink was removed to form a lumen when collagen bioinks were incubated at 37 °C, as shown in **Figure 1C**. The widths of the surface area, side area, and lumen area in the printed construct were 36.45 ± 13 , 51.66 ± 28 , and 98.21 ± 8.7 μm, respectively (**Figure 1B**). Although we designed and fabricated a 1 mm inner diameter of the nozzle to mimic the size of a hepatic lobule, smaller nozzles can be exploited to improve printing resolution. Nevertheless, our printing resolution (10 μm) was shown to be higher than conventional extrusion-based bioprinting (100 μm). Next, we assessed the coprintability of bioinks using collagen solutions with alginate solutions (sacrificial material) and alginate solutions with gelatin solutions (sacrificial materials). The coprintability of the bioink was achieved by increasing each group of bioinks. Coprintable concentration of the collagen bioinks with alginate solution was each 3–5%, and alginate with gelatin was 2–4% when gelatin remained constant at 35% (**Figure 1D–I**). Low concentration of alginate solution or collagen solution could not be coprinted due to low viscosity of the materials. Gelatin was unsuitable for preset extrusion bioprinting due to thermal sensitivity, which affected the structural integrity (**Figure S1**, Supporting Information). Thus, in this study, we used the combination of 3% collagen solution and 3% alginate solution.

2.2. Hepatic Lobule Vasculature

A vascular system is essential for survival of humans and larger tissues constructs.^[46] Despite recent advances in bioprinting, vascularization remains an elusive objective, which limits effective in vivo translation and scale-up bioprinting.^[21] Alternatively, coaxial nozzle and sacrificial materials have been utilized to create microchannels. This study utilized the sacrificial material, alginate, to build a microchannel. We hypothesized

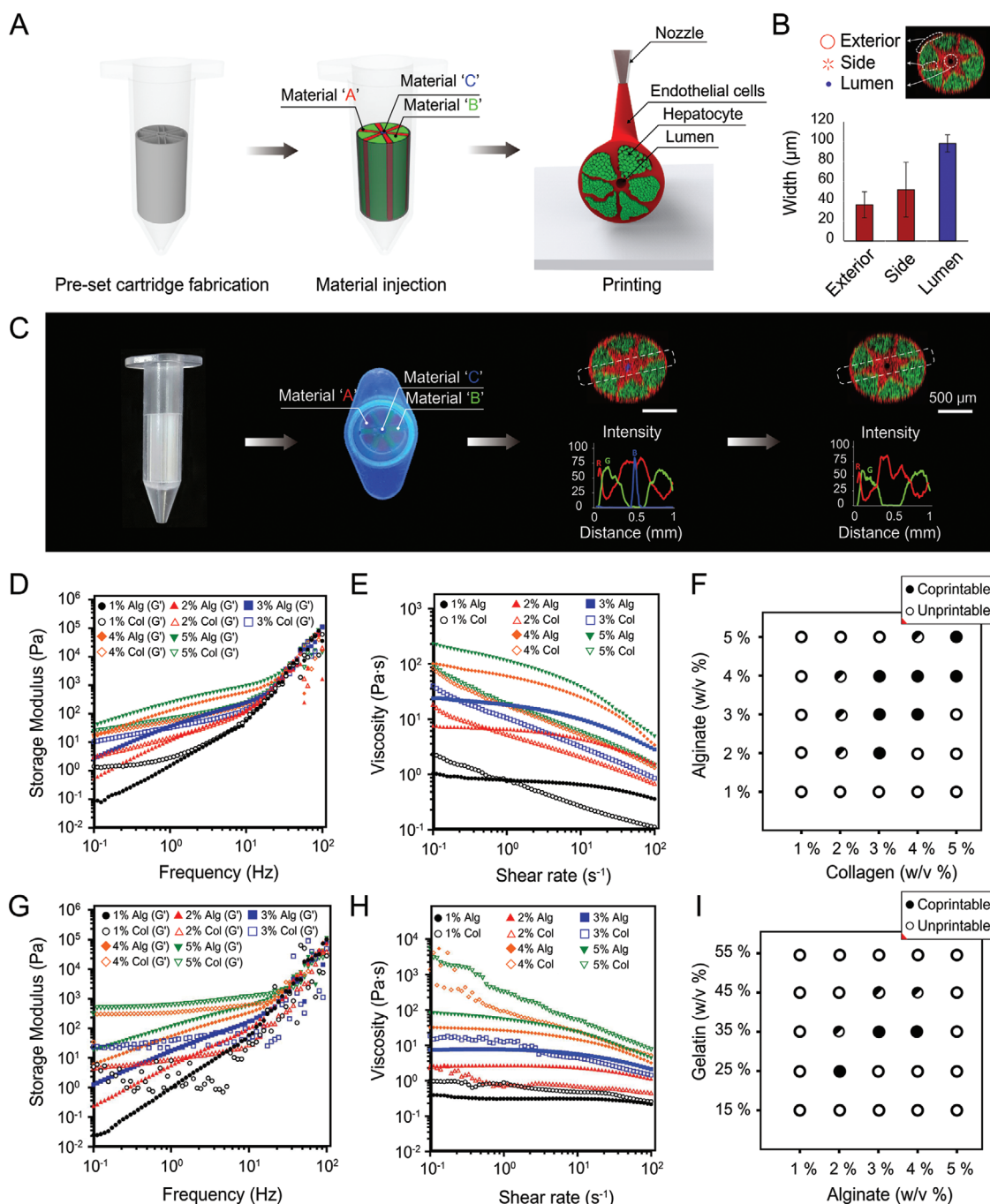


Figure 1. Experimental design and demonstration of preset extruded hepatic lobule. A) Schematic illustration of preset extrusion bioprinting technique for hepatic lobule printing. B) Average width of the printed construct. C) Macroscale of preset cartridge, microscale of printed construct, and its average intensity profile. D–I) Rheological analysis of bioinks and coprintability of each bioink combination (collagen with alginate and alginate with gelatin) ($n = 3$).

that adjacent ECs at the lumen or surface could form a monolayer and simultaneously interconnect ECs located between the lumen and exterior. To visualize this, we fabricated hepatic lobule models without HepG2/C3A, in which the areas were filled with hydrogel only. Subsequently, we observed the surface area (i.e., exterior), lumen area, and side area (i.e., gap between the surface and lumen) for 7 days (Figure 2A). The ECs viability

was slightly higher in the surface area, whereas the lumen area was gradually decreased. Nevertheless, cell viability was maintained at greater than 90% for 7 days (Figure 2B,C). ECs covered both the surface and lumen area as expected on day 7 (Figure 2B,D). Regarding the morphological changes, ECs were stationary in both the surface area and lumen area for 7 days, thus the aspect ratio was kept closely to 1. It indicated

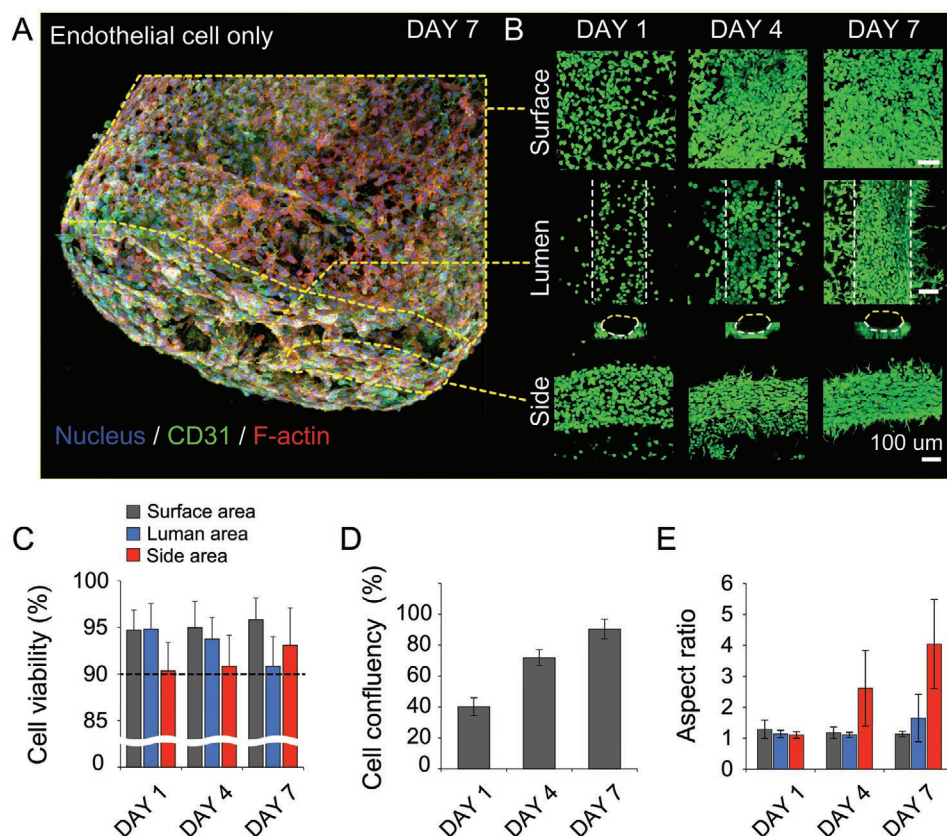


Figure 2. Bioprinted hepatic lobule without hepatic cell. A) Immunostaining of CD31 (green), f-actin (red), and nucleus with DAPI (blue). B) Live cells stained with calcein AM (green) and dead cells stained with ethidium homodimer-1 (red) to visualize morphological changes and cell viability. C) Cell viability, D) cell confluency, and E) aspect ratio of endothelial cells at the surface, lumen, and side area.

that ECs grew on the surface and lumen area as flat 2D cultured ECs. In contrast, ECs in the side area were gradually elongated, and subsequently the aspect ratio changed $\approx 1-4$ for 7 days (Figure 2E). As a result, ECs grew confluent in bioinks and elongated in the side area. Remarkably, ECs could interconnect together in most areas, except the hepatic area (Figure 2A).

The survival of the engineered tissue construct following *in vivo* transplantation depends on the rapid development of vascular network throughout the implant.^[47] Thus, *in vitro* prevascularization means seeding ECs with or without vascular support cells into scaffolds or hydrogels *in vitro* with subsequent implantation *in vivo*.^[48] To increase graft–host vessel anastomosis, the presence of ECs in the implanted engineered tissue is essential. A recent study has proved that implantation of the engineered graft showed no significant differences in graft perfusion compared to the early stage of the prevascularized scaffold and non-prevascularized scaffold. However, a significant higher perfusion rate was observed in grafts bearing mature vessel.^[49] Therefore, the presence and maturation of ECs were both important prior to an engineered tissue implant.

2.3. Structural Integrity of Hepatic Lobule Model

Next, we assessed structural integrity of the hepatic lobule model by observing cell viability and whole structures for

10 days. We prepared three models: mixture of HepG2/C3A and ECs without lumen (Mix), mixture of HepG2/C3A and ECs with lumen (Mix (L)), and preset extruded hepatic lobule model (Preset). As previously demonstrated, our hepatic lobule printing techniques enabled spatial arrangement of cells heterogeneously, creation of the lumen, and formation of a liver lobule-like structure after bioprinting. As shown in Figure 3A, each group showed high cell viability at day 1 (Figure 3C), whereas dead cells gradually increased on day 5 in the nonengineered model (Mix) compared to the lumen-embedded models (Mix (L), Preset). Further, proliferated cells began to decrease on day 3 in Mix (Figure 3B) due to diffusional limitation of oxygen and nutrients for cells located at the center. Remarkably, centrally located cells did not survive, as shown in the inset of Figure 1A for day 5; thus, the cells migrated to the edge or grew at the edge of Mix (Figure 3A, Day 10). Although lumen embedded models maintained high cell viability at day 5, the lumen was gradually filled with a mixture of cells in the Mix model (Figure S2, Supporting Information), which resulted in relatively lower cell viability and proliferation (Figure 3B,C). As a result, the Mix model showed relatively lower cell viability and proliferation compared to the lumen-embedded Mix model due to the pre-existing lumen. Likewise, the lumen-embedded Mix model showed relatively lower cell viability and proliferation compared to the Preset model due to spatial cell arrangements, which resulted in reduced circularity and diameter of the lumen

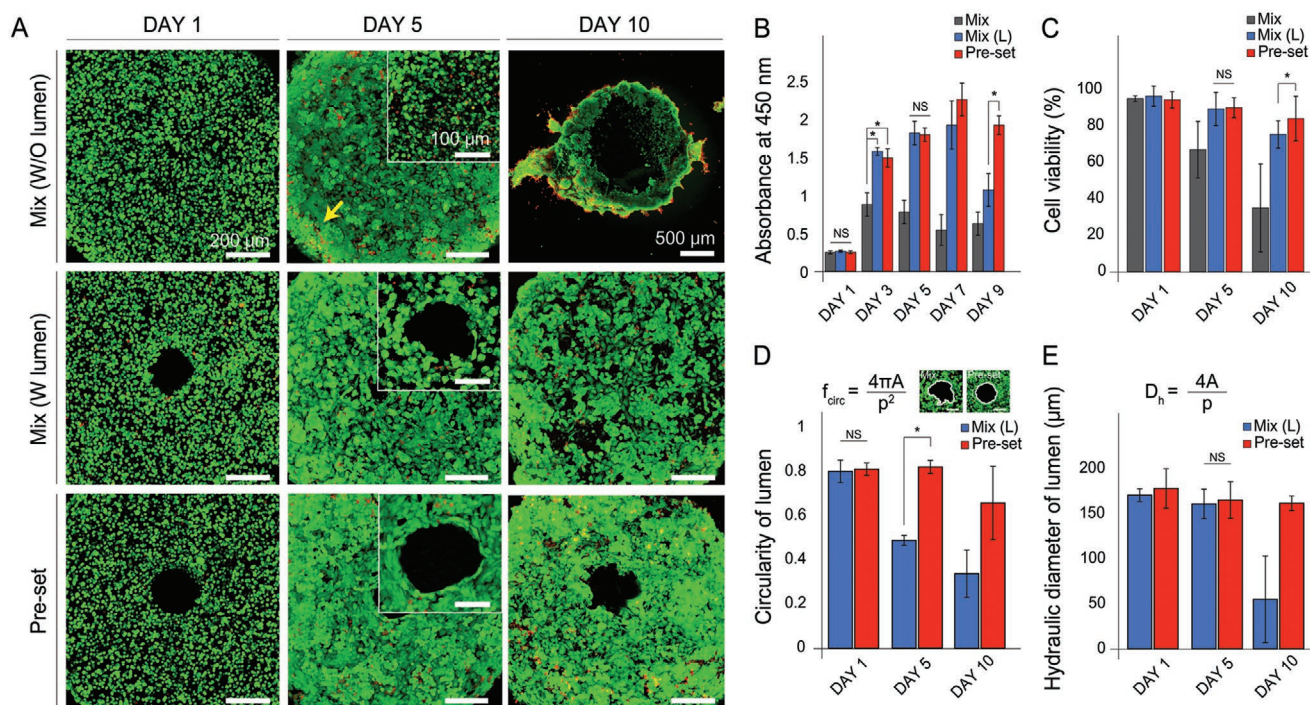


Figure 3. Structural integrity of printed constructs. Mix (W/O lumen) represents HepG2/C3A/ECs cell mixtures without lumen, Mix (W lumen) is HepG2/C3A/ECs cell mixtures with lumen, and Preset is the preset extruded hepatic lobule. A) Live cells stained with calcein AM (green) and dead cells stained with ethidium homodimer-1 (red) to observe structural integrity of each model. B) Cell proliferation and C) viability of each model. D) Circularity and E) hydraulic diameter of lumen in Mix (W lumen) and Preset groups. “*” indicates statistical significance ($p < 0.05$), and NS indicates nonsignificance ($n = 10$).

(Figure 3D,E). Thus, in engineering a complex tissue such as a liver, each of the various cell types should be spatially arranged in a pattern similar to normal human liver tissue.

2.4. Hepatic Function

To assess hepatic function, Mix (with lumen) and Preset models were stained with albumin and MRP2, including CD31 for ECs. As shown in Figure 4A, structural integrity, including endothelium-lined lumen, was well preserved in the Preset but not in the Mix group. Because spatial cell arrangement enhanced cellular organization of each cell type, Mix showed structural inconsistencies. Specifically, ECs covered the surface of the structure, which surpassed the space of HepG2/C3A, and both cells were not maintained in the central lumen of Mix. Further, we analyzed albumin secretion and urea production in Mix, Preset, and 2D (i.e., HepG2/C3A only). Although the 2D culture model could not survive for 7 days, the produced urea was similar to the other models. However, secreted albumin was relatively lower than the other models on days 1 and 4. Specifically, albumin levels were gradually increased in Preset on day 7, whereas albumin decreased in Mix at day 7. Similar results were observed for urea production. Thus, although urea production was slightly decreased in Preset, both albumin secretion and urea production were gradually increased or stably maintained on day 7 (Figure 4D). Subsequently, western blot analysis (Figure 4B) showed that protein levels of albumin and MRP2 were gradually increased on day 4 and then decreased on

day 7 in both models. However, overall albumin and MRP2 protein levels were higher in Preset compared to Mix. In addition, MRP2 expression was significantly higher in Preset on day 7 due to a rapid decrease in Mix (Figure 4C). CD31 expression was also significantly increased in both models and higher in Preset compared to Mix. Further, we assessed enzyme activity of CYP3A4 induced by rifampicin (50×10^{-6} M) and CYP1A1 induced by omeprazole (25×10^{-6} M) on day 8. For 2D culture of HepG2/C3A, P450 enzyme activity was measured at day 5 due to relatively lower cell viability. The results showed both enzyme activities of Preset and Mix were 2 times higher than 2D. Preset showed higher enzyme activity of CYP3A4 and CYP1A1 compared to Mix (Figure 4E). To further assess hepatotoxicity test, the sample (2D, Mix, Preset) was treated with amiodarone, which is an antiarrhythmic medication that causes hepatotoxicity by affecting mitochondrial function,^[50] for 2 days at the following concentrations: 250×10^{-6} , 100×10^{-6} , 80×10^{-6} , 60×10^{-6} , 40×10^{-6} , and 20×10^{-6} M. Control cells were treated with 1% DMSO in culture medium. The result showed that cell viability of the samples was gradually decreased as the amiodarone concentration increased. However, administration of low-dose amiodarone from 40×10^{-6} to 60×10^{-6} M resulted in different cell viability between 2D and 3D models. Specifically, the viability of 2D was lower than 50%, but not in Preset and Mix. We found that the drug resistance or sensitivity between 2D and 3D differed due to dimensionality and the presence or absence of ECM and ECs.^[51] Mix was inappropriate to measure hepatic toxicity due to the largest value of standard deviation (SD). Subsequently, as shown in

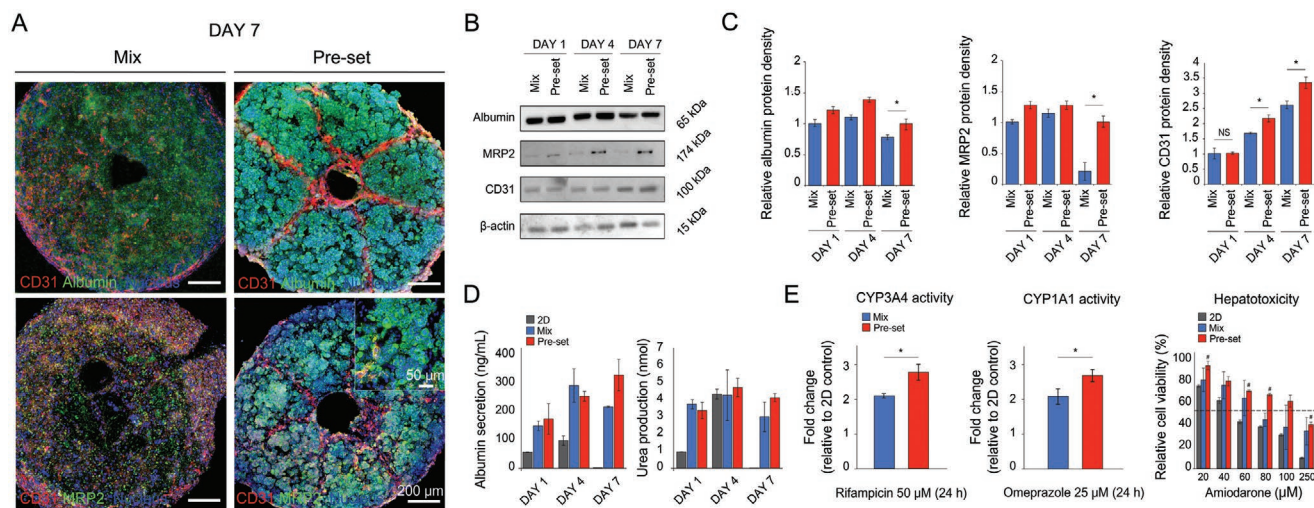
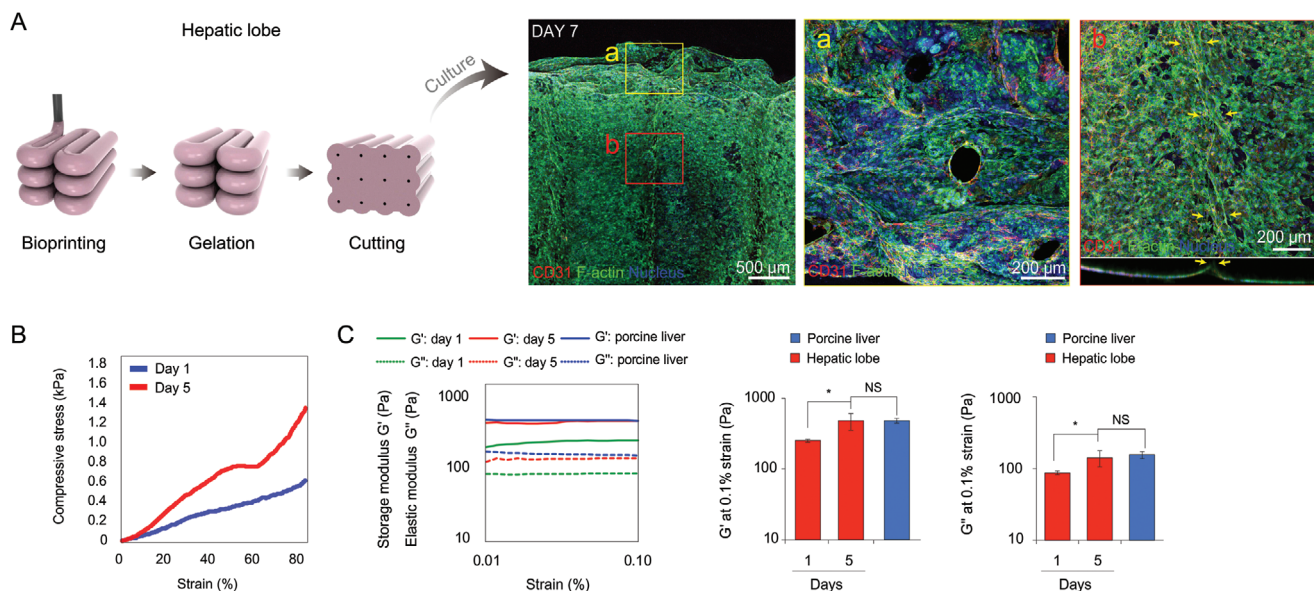


Figure 5A, we obtained multiscaled hepatic lobes by layer-by-layer deposition of hepatic lobules. The schematic illustration of the bioprinting process for hepatic lobes and a macroscopic view of hepatic lobes are shown. The dimension of bioprinted hepatic lobes was $\approx 4 \times 5.2 \times 2.5$ (mm) (Figure S4, Supporting Information). Sequentially, low magnification view of hepatic lobes showed stacked layers (Figure S3, Supporting Information; three-layers) and the struts. The high magnification view showed hepatic lobule arrays and endothelialization between struts. To evaluate the mechanical property of hepatic lobes,

compression test and rheological analysis were performed. The strain–stress curve showed that mechanical strength of hepatic lobes was increased on days 1–5 (Figure 5B), and both Young’s modulus and maximum compressive strength were increased on day 5 compared to day 1, more than two times (Figure S5, Supporting Information). To demonstrate scalability of hepatic lobes, both storage modulus and elastic modulus were evaluated in the hepatic lobe compared to porcine liver (Figure 5C). The result showed that storage modulus and elastic modulus were significantly increased on days 1–5 in the hepatic lobe.



Finally, the hepatic lobe on day 5 had similar viscoelastic property to porcine liver.

In this study, we used two immortalized cell lines (HepG2/C3A and EA.hy926) with the preset extrusion bioprinting technique. Our findings demonstrated that the preset extrusion bioprinting technique enabled spatial arrangement of cells heterogeneously, creation of the lumen, and formation of a liver lobule-like structure. Moreover, we revealed that each cell type with spatial cell patterning in bioink accelerated cellular organization, which could preserve structural integrity and improve cellular functions. Further study showed that macroscale hepatic lobes (i.e., hepatic lobule arrays) could be printed with similar mechanical property to the porcine lobe. However, the use of immortalized cell lines is a limitation for hepatic study. Since primary human hepatocytes are difficult to seek and culture, alternative cell lines are still needed. To supplant the limited primary hepatocyte culture, many researchers have sought liver stem cells, which are believed to reside in biliary ducts. They found that cholangiocytes (bile ductal cells) regenerated hepatocytes during severe liver injury.^[52] Lin et al. have shown that a subset of hepatocytes expressed high levels of telomerase and are expressed in many stem cell populations, which aids in the repopulation of liver cells during hemostasis and injury.^[53] Huch et al. have reported that LGR5+ adult bile duct-derived biopotent progenitor cells from human liver can be converted into functional hepatocytes in vitro and upon transplantation in vivo.^[52] Further, liver organoid technology is strongly dependent upon endogenous cell organization, self-assembly or self-organization, with xenogeneic components, such as Matrigel. Hence, several researchers have developed organoid models under human extracellular matrix (ECM)-based conditions or using human induced pluripotent stem cells (iPSC) to expand the use of organoids for clinical trials. For instance, Takebe et al. have developed an omni-well-array culture platform for mass production of homogeneous and miniaturized liver buds for clinical applications. Specifically, they have used iPSC-derived hepatic endoderm, endothelium, and septum mesenchyme, which enables self-condensation, hepatic maturation, and vascularization of a successful in vivo transplant.^[54] In this context, we aim to use an iPSC-derived hepatocyte, which is one of the suitable substitutes of HepG2/C3A, hepatic sinusoidal endothelial cells, hepatic stellate cells, and Kupffer cells or immune cells, as well as primary cells, for hepatic function and tissue-engineered grafts.

3. Conclusion

Here, we fabricated hepatic lobule models that embedded a high cell density of both HepG2/C3A and ECs in a hepatic lobule structure (1 mm), including a central lumen (150 μ m), using the preset bioprinting technique. The well-organized ECs covered an exterior of the constructs for structural integrity, formed a lumen, and finally interconnected from the lumen to the exterior of the construct. The engineered hepatic lobules showed stable secretion of albumin and urea, and higher protein levels of albumin, MPR2, and CD31 compared to non-engineered models. Therefore, high cell density with spatial cell

patterning in bioink accelerated cellular organization by adjacent cell–cell and cell–ECM interaction, which can preserve structural integrity and improve cellular functions. Further studies showed that the complex heterogeneous hepatic lobes as a series of lobules can be printed using the preset extrusion bioprinting technique, which enables micro- to macro-scale tissue engineering.

4. Experimental Section

Design and Fabrication of Preset Cartridge: In this study, the preset cartridge was utilized, in which a precursor cartridge was well defined in the previous study. For human scale hepatic lobule bioprinting, a hepatic lobule cartridge was designed using SolidWorks software (Dassault Systems, France). Subsequently, the 3D CAD file was converted to STL-files, and the STL-files were imported into the 3D printing system software for rapid prototyping (Projet 6000, 3D systems). The fabricated preset cartridge (Visijet SL clear, 3D systems) was sterilized with 70% ethanol for 10 min. Following, ultraviolet light was applied for 6 h, and the cartridge was soaked in culture medium overnight before use.

Bioinks: Sodium alginate (viscosity > 2000 cP, 25 °C, Sigma-Aldrich) in Dulbecco's phosphate-buffered saline (Gibco) was prepared and stirred for 6 h at room temperature. Concentrations of the fabricated alginate solution ranged from 1 to 5 w/v%, which was used for both cell-laden bioinks and sacrificial materials.

Lyophilized Atelo-collagen derived from the porcine tendon was purchased from MS bio (Republic of Korea). A collagen solution (pH 4.0) was mixed with 10 $\times$ Dulbecco's modified Eagle medium (DMEM) solution and reconstituted buffer (2.2 g NaHCO₃, 4.77 g hydroxyethyl piperazine ethane sulfonic acid, 0.2 g NaOH in 100 mL distilled water) at a volume ratio of 8:1:1. Finally, the neutralized collagen was obtained from 1 to 6 w/v%.

Gelatin powder (type A, Sigma-Aldrich) was dissolved in distilled water at 50 °C for 1 h to achieve a final concentration of 15, 25, 35, 45, or 55 w/v% for the gelatin solutions.

Cell Culture: HepG2/C3A (HB-8065, ATCC) and EA.hy 926 (CRL-2922, ATCC) cell lines were cultured in low glucose DMEM (Gibco) supplemented with 10% fetal bovine serum (Gibco) and 1% penicillin/streptomycin (Gibco) at 37 °C under 5% CO₂. Both cells were detached from the cell culture dish (Thermo Fisher Scientific) using TrypLE (Gibco) and then blended with prepared bioinks before bioprinting.

Rheological Analysis: Rheological properties of the various bioinks were measured using a Kinexus Pro + rheometer (Malvern, Worcestershire, UK) equipped with a geometric cone plate with a 20 mm diameter (PU20 SR3650 SS) and 1 mm gap. To prevent collagen gelation, rheological analysis of collagen was performed under 4 °C, which was the same as the printing temperature. Likewise, to prevent gelatin gelation, rheological analysis was performed at 37 °C, which was the same as the printing temperature. To optimize coprinting conditions, coprintability analysis was performed between collagen and alginate solutions, as well as alginate and gelatin solutions. Coprintable or unprintable conditions were determined from 3D z-staked images, in which the preset cartridge with fluorescent particle-laden bioinks was printed. The representative unprinted and coprinted images are shown in Figure S1 in the Supporting Information. Subsequently, to analyze rheological properties of the hepatic lobe, porcine liver was cut to $\approx 4 \times 5.2 \times 2.5$ mm, which were the dimensions of the hepatic lobe. Both storage modulus and elastic modulus (e.g., rheological properties) on the hepatic lobe and porcine liver were measured as described above.

Mechanical Property: The compressive strength of the sample was tested using a universal testing machine Instron 3345 at room temperature with a crosshead speed of 1 mm min⁻¹. The hydrogels and porcine liver were gently swollen in distilled water to prevent evaporation. Sample dimension was $\approx 4 \times 5.2 \times 2.5$ mm. Young's modulus was evaluated from the slope of the strain–stress curve.

Image Analysis: To evaluate cell viability, live and dead cells were stained with Calcein AM and ethidium homodimer-1, respectively (LIVE/DEAD Viability/Cytotoxicity Kit, L3224, Thermo Fisher Scientific).

The percentage of cell confluency was evaluated using Image J analysis to quantify cell spreading area and whole area. Subsequently, the single-cell aspect ratio was assessed using Image J analysis to quantify the ratio of the longer side to its shorter side (the ratio of width to height at a right angle) of a single cell.

A methodology for both circularity and hydraulic diameter of the lumen is marked as an inset of Figure 3B,C. At least 10 randomly selected areas of each sample were used for image analysis using Image J software. The methods of image analysis have been described in the previous studies.^[15]

Bioprinting: To bioprint hepatic lobules, HepG2/C3A and ECs were manually mixed with neutralized collagen (each 6×10^7 cells per mL) on ice. Following, the bioink (i.e., cell-laden neutralized collagen) was centrifuged at 4000 RPM for 10 min to remove any bubbles. Subsequently, bioinks containing alginate (3 w/v%) were injected into a precursor cartridge, and the bioink-laden precursor cartridge was inserted into 3DX bioprinter (T&R Biofab, Korea). Finally, the bioink was extruded into prewarmed culture medium on a thermal plate and kept immersed for 1 h in an incubator at 37 °C and 5% CO₂. To remove sacrificial material (i.e., alginate), the sample was cut by surgical blade and moved to a 24-well culture plate. The diameter of the bioprinted constructs was ≈ 1 mm, and the thickness was ≈ 500 μ m.

Likewise, to fabricate hepatic lobes (hepatic lobule arrays), the bioink was extruded into prewarmed culture medium and kept immersed for 1 h in an incubator as described above. To remove sacrificial material (i.e., alginate), the sample was cut by surgical blade as shown in Figure S4 in the Supporting Information and moved to a 100 mm cell culture plate. The fabricated hepatic lobes were printed with a feed rate of 400 mm min⁻¹ at 45 kPa using a nozzle with a 500 μ m inner diameter. The construct was composed of a 3 $\times$ 4 strut array, with each strut having a central lumen. The overall width of construct was 4 mm, with a height of 5.2 mm and thickness of 2.5 mm.

Albumin Secretion and Urea Synthesis: For evaluating hepatic functions, each group of cell culture supernatants was collected for 1, 4, and 7 days and stored at -20 °C before use. Albumin secretion was assessed using the human albumin in vitro ELISA (Enzyme-Linked Immunosorbent Assay) kit (ab108788, Abcam), and urea production was measured using a Urea Assay Kit (ab83362, Abcam) according to the manufacturer's instructions. The cultured medium was changed every day, and three samples each were grown together in one well of a 12-well plate.

Immunofluorescence Analysis: Mix and Preset models were stained with albumin and MRP2 to verify hepatic function and stained with CD31 for the hepatic lobule vasculature, which can distinguish each cell type. In brief, the washed samples were fixed in 4% formaldehyde solution for 5 min and rinsed with phosphate-buffered saline (Gibco) five times. For permeabilization, the samples were soaked with 0.25% Triton-X 100 for 10 min and blocked with 1% bovine serum albumin (BSA, GenDEPOT) solution for 1 h. The samples were immunostained using a primary antibody against albumin (1:100, ab10241, Abcam) or MRP2 (1:50, ab3373, Abcam) overnight. Subsequently, the samples were incubated with Alexa Fluor 488 goat anti-mouse (1:500) for albumin and MRP2 overnight. Sequentially, the samples were immunostained using a primary antibody against CD31 (200:1, ab76533, Abcam) overnight, followed by an overnight incubation using Alexa Fluor 594 goat anti-rabbit (1:500). Finally, the samples were counterstained with 4',6-diamidino-2-phenylindole (DAPI, blue) and visualized using a laser-scanning confocal microscope (Olympus). Likewise, the HepG2/C3A-free sample was stained with CD31 as described above, but counterstained with DAPI and phalloidin (red).

Enzyme Activity: The CYP3A4 and CYP1A1 activity were evaluated using the P450-Glo CYP3A4 and CYP1A1 assay kit (Promega). Each sample was incubated with culture medium containing rifampicin (50×10^{-6} M) for the CYP3A4 activity assay and omeprazole (25×10^{-6} M) for the CYP1A1 activity assay. The culture medium containing DMSO

(0.1×10^{-6} M) was utilized as a control group. The activity of each enzyme was measured using a GloMax navigator microplate luminometer (Promega) according to the manufacturer's instructions. Sequentially, to assess hepatic toxicity, the sample was treated with amiodarone for 2 days at the following concentrations: 250×10^{-6} , 100×10^{-6} , 80×10^{-6} , 60×10^{-6} , 40×10^{-6} , and 20×10^{-6} M. Control cells were treated with 1% DMSO in culture medium. The cell viability was evaluated using the CellTiter-Glo 2.0 Cell Viability Assay (Promega) and measured by the microplate luminometer.

Western Blot: Protein extracts, separated by sodium dodecyl sulfate polyacrylamide gel electrophoresis and transferred to a polyvinylidene difluoride (PVDF) membrane, were probed with primary antibodies against CD31 (1000:1, Cell Signaling, #3528), albumin (500:1, Abcam, ab106582), MRP2 (500:1, Abcam, ab110740), and beta actin (1000:1, Cell Signaling, #3700 or 1000:1, Cell Signaling, #4970). The proteins were detected with Goat anti-Rabbit IgG (H+L) Cross-Adsorbed Secondary Antibody (1500:1, Thermo Fisher Scientific, G-21234) or Goat anti-Mouse IgG (H+L) Cross-Adsorbed Secondary Antibody (1500:1, Thermo Fisher Scientific, G-21040). The protein levels were visualized with the ChemiDoc touch imaging system (Bio-rad).

Statistical Analysis: All values are represented as mean $\pm$ SEM, whereas image analysis is represented as mean $\pm$ SD. Mann-Whitney U test (two groups) and the Kruskal-Wallis test with Tukey's post-hoc test (more than two groups) were used. $p < 0.05$ was considered statistically significant. NS is used to denote "no significance." At least three individual experiments were performed for data collection. The statistical software SPSS 18.0 (SPSS Inc.) and GraphPad Prism 7.0 (GraphPad Software Inc.) were used for data analysis.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

bioprinting, hepatic lobules, liver, liver tissue fabrication, tissue engineering

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